Module -01

1. **Explain the basic architectural elements** of a Wireless Sensor Network (WSN) with a neat diagram.
2. **Discuss the main advantages and challenges** faced while designing and deploying WSNs.
3. **Write a short note on the applications** of WSNs in environmental monitoring and military surveillance.
4. **Explain sensor node technology** by describing the major hardware components of a standard sensor node.
5. **Describe sensor taxonomy** in WSNs. How are sensors classified based on energy consumption and data source?
6. **Explain the impact of the WSN operating environment** on wireless communication and node deployment.
7. **Discuss the radio technology options** used in WSNs along with their constraints.
8. **Explain the primary optimization goals** and figures of merit when structuring a WSN.
9. **List and describe the design principles** that must be followed for effective WSN deployment.
10. **Write a short note on Gateway concepts** in WSNs and how they connect sensor fields to external networks.
11. **Provide an overview of Wireless Sensor Network Operating Systems** and explain why traditional OS designs are not suitable for sensor nodes.
12. **Explain the features and architecture of TinyOS** as a case study for WSN operating systems.
13. **What is an Ad-hoc Network?** Discuss its role and implementation inside a WSN infrastructure.
14. **Differentiate between MANETs (Mobile Ad-hoc Networks) and WSNs** based on deployment, scale, and energy constraints.
15. **Discuss the energy efficiency considerations** that must be taken into account in WSN ad-hoc routing.
16. **Explain the main security and privacy issues** encountered in WSN ad-hoc network environments.

17. **Explain the fundamentals and primary goals** of MAC protocols designed specifically for wireless sensor networks.
18. **Identify the major sources of energy waste** at the MAC layer (e.g., collision, idle listening, overhearing) and how they are addressed.
19. **Explain the S-MAC (Sensor-MAC) protocol** in detail, highlighting its periodic sleep and wake-up mechanisms.
20. **Discuss the concept of collision avoidance and synchronization** as implemented in the S-MAC protocol case study.
21. **Elaborate on the key routing challenges** and design issues unique to WSN environments.
22. **Compare data-centric routing with hierarchical routing** strategies in WSNs.
23. **Write a detailed note on the IEEE 802.15.4 LR-WPAN standard** and explain its importance to WSN operations.
24. **Explain how data dissemination and gathering work** as basic routing strategies in sensor fields.
25. **Why do traditional transport control protocols (like standard TCP/UDP) fail** when applied directly to WSNs?
26. **Discuss the key design issues and performance metrics** when designing a WSN Transport Layer protocol.
27. **Explain the concept and architectural role of WSN middleware**. How does it isolate application logic from hardware constraints?

Module -02

- **Explain the different types of signal propagation behaviors** (such as reflection, diffraction, and scattering) in wireless mediums.
- **Differentiate between Multiplexing and Modulation**. Why are both essential for wireless transmission?
- **Write a short note on Space Division Multiplexing (SDM) and Frequency Division Multiplexing (FDM)**.
- **Explain the concept of Spread Spectrum**. Briefly compare **DSSS** (Direct Sequence) and **FHSS** (Frequency Hopping).
- **Discuss the core concept of Cellular Systems**, highlighting the significance of frequency reuse and cell splitting.
- **Explain the functional architecture of a GSM system** with a neat, labeled block diagram showing BSS, NSS, and OSS.
- **Write a short note on GSM mobile services** (Bearer services, Tele-services, and Supplementary services).
- **Describe the GSM Radio Interface**. Explain how FDMA and TDMA are combined in GSM structures.
- **Explain the step-by-step process of Handover** in GSM. Why is it necessary during an active call?
- **Discuss the localization and calling mechanism** in GSM. How do HLR and VLR assist in tracking a mobile station?
- **Explain the security mechanisms used in GSM** (Authentication, Encryption, and IMSI confidentiality).
- **Write a short note on new data services in GSM**, specifically highlighting **GPRS** and **EDGE**.
- **Explain the system architecture and protocol architecture of DECT** (Digital Enhanced Cordless Telecommunications).
- **Write a short note on TETRA (Terrestrial Trunked Radio)** and its typical industrial/emergency applications.
- **Discuss the features and architecture of UMTS (Universal Mobile Telecommunications System)** as a 3G technology.
- **Explain the key objectives and requirements defined by the IMT-2000** standard framework.

- **Compare GEO, LEO, and MEO satellite orbits** based on their altitude, round-trip delay, and coverage area using a quick comparison table.
- **Discuss the major applications of modern satellite systems** in communication, navigation, and remote sensing.
- **Explain the challenges and strategies involved in Routing** within satellite-to-satellite networks.
- **Describe how Localization and Handover occur** in satellite communication networks when a user moves across footprints.